
Running V FOODS Locally

A practical step-by-step setup guide for running the V FOODS frontend and backend on your local computer for development and testing.

1. Prerequisites

Before starting, make sure you have the following installed on your computer:

- **Node.js:** Version 18 or newer (comes with `npm`).
 - **Python:** Version 3.10 or newer (comes with `venv` and `pip`).
 - **Git:** For pulling the code and managing branches.
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2. External Service Dependencies

V FOODS connects to three cloud services. To run the full system with live data, you will need credentials from each:

1. **Supabase Project:** Provides the PostgreSQL database, authentication, Row-Level Security, and Realtime WebSocket events.
 2. **Paytm Merchant Credentials:** Provides sandbox/staging gateway credentials (`PAYTM_MID` , `PAYTM_MERCHANT_KEY` , `PAYTM_WEBSITE` , `PAYTM_CHANNEL_ID` , `PAYTM_INDUSTRY_TYPE_ID`) for testing wallet top-ups and direct order checkout with mock UPI/Cards.
 3. **MSG91 Account (Optional in Local Dev):** Provides SMS and WhatsApp messaging APIs for order ready notifications. If left blank, the app logs notifications to the console instead of sending live SMS.
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3. Environment Variables Setup

Environment variables are configuration settings that tell the app where to connect without exposing private passwords in the code.

Frontend Configuration

Create a file named `.env` inside the `frontend/` directory with these variable names:

- `VITE_SUPABASE_URL` : The web address of your Supabase project (e.g. `https://xyzcompany.supabase.co`).
- `VITE_SUPABASE_ANON_KEY` : The public anonymous API key provided in your Supabase project settings.

- `VITE_API_URL` : The local web address of your running FastAPI backend (typically `http://localhost:8000`).

Backend Configuration

Create a file named `.env` inside the `backend/` directory with these variable names:

- `SUPABASE_URL` : The web address of your Supabase project.
- `SUPABASE_SERVICE_ROLE_KEY` : The private service role key from Supabase (used by the backend for admin operations).
- `PAYTM_MID` : Your Paytm Merchant ID (use staging MID for testing).
- `PAYTM_MERCHANT_KEY` : The private cryptographic AES/HMAC merchant key provided by Paytm.
- `PAYTM_WEBSITE` : `WEBSTAGING` (for test mode) or `DEFAULT` (for production).
- `PAYTM_CHANNEL_ID` : `WAP` (for mobile web checkout) or `WEB` .
- `PAYTM_INDUSTRY_TYPE_ID` : `Retail` .
- `PAYTM_ENV` : `STAGE` (for sandbox testing) or `PROD` (for live).
- `WEBHOOK_BASE_URL` : The public HTTPS address where Paytm sends webhook callbacks (e.g. Ngrok tunnel or cloud host URL).
- `FRONTEND_URL` : The address of the frontend app (e.g. `http://localhost:5173`).
- `MSG91_AUTH_KEY` : Your MSG91 authorization key for messaging (optional in dev).
- `MSG91_WHATSAPP_INTEGRATED_NUMBER` : The verified WhatsApp sender number registered on MSG91.
- `QR_SECRET` : A random secret passphrase used to generate secure HMAC-signed QR codes for order collection.

4. Starting the Frontend (React + Vite)

1. Open your terminal and navigate to the `frontend` folder: `bash cd frontend`
2. Install the necessary project dependencies: `bash npm install`
3. Start the local development server: `bash npm run dev`
4. The website will open in your browser at `http://localhost:5173` . You can log in using test credentials or create a new student account.

5. Starting the Backend (FastAPI Python)

1. Open a second terminal window and navigate to the `backend` folder: `bash cd backend`
2. Create and activate a Python virtual environment:
3. On Windows: `bash python -m venv venv .\venv\Scripts\activate`
4. On macOS / Linux: `bash python3 -m venv venv source venv/bin/activate`
5. Install required Python packages: `bash pip install -r requirements.txt`

6. Start the FastAPI server: `bash uvicorn main:app --reload --port 8000`
 7. The API will start at `http://localhost:8000` . You can test endpoints directly in your browser using the built-in documentation at `http://localhost:8000/docs` .
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6. Database Migrations Setup

1. Log in to your Supabase web dashboard.
 2. Open the **SQL Editor**.
 3. Run the SQL migration scripts located in the `supabase/migrations/` folder in sequential order (`001_...` , `002_...` , etc.).
 4. This will create all required tables (`outlets` , `menu_items` , `orders` , `wallets` , etc.), configure Row-Level Security rules, and install the atomic wallet procedures.
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